

Lab Worksheet - Exploratory Data Analysis in airquality

Dataset

```
# load libraries:
library(tidyverse) # Recall that tidyverse contains dplyr, ggplot2, and many
other useful packages.

# Load 'airquality' dataset:
data("airquality")
airquality |>
  head()
```

	Ozone	Solar.R	Wind	Temp	Month	Day
1	41	190	7.4	67	5	1
2	36	118	8.0	72	5	2
3	12	149	12.6	74	5	3
4	18	313	11.5	62	5	4
5	NA	NA	14.3	56	5	5
6	28	NA	14.9	66	5	6

Q1: Render this quarto file to a pdf. To do so, you can use the hotkey **Ctrl + Shift + K** or **Cmd + Shift + K**. Then, you will see a command appear in the Positron Terminal. Once the command finishes running, you will see a pdf file appear in the same directory as this file and a **VIEWER** window will display the pdf on the right pane of Positron. Here you should see the rendered contents of this file.

Q2: Convert the **Month** column to a factor with it's levels labeled as month names (e.g. "May", "June", etc.).

```
# your code here
```

Q3: Create a new column **Temp_C** that converts the temperature from Fahrenheit to Celsius.

```
# your code here
```

Q4: Rename the **Wind** column to **Wind_mph** and the **Temp** column to **Temp_F**.

```
# your code here
```

Q5: Create a scatter plot of `Ozone` vs `Temp_C`, colored by `Month`. Add appropriate axis labels and a title.

```
# your code here
```

Q6: Create a violin plot of `Ozone` levels for each month. Add appropriate axis labels and a title.

```
# your code here
```

Q7: Create a segmented bar chart of the proportion of days with `Ozone` levels above 50 by `Month`. Remove NA values in the `Ozone` column. Add appropriate axis labels and a title.

```
# your code here
```

Q8: Create a stacked bar chart showing the count of days available for each month. Fill bar colors by `Ozone` levels above or below 50 for each month. Remove NA values in the `Ozone` column. Add appropriate axis labels and a title.

```
# your code here
```

Q9: What is the utility of using a segmented bar chart vs a regular bar chart in this context? Which do you think is more informative and why?

your answer here

Q10: Create a summary table of the number and proportion of NA values in each column.

```
# your code here
```